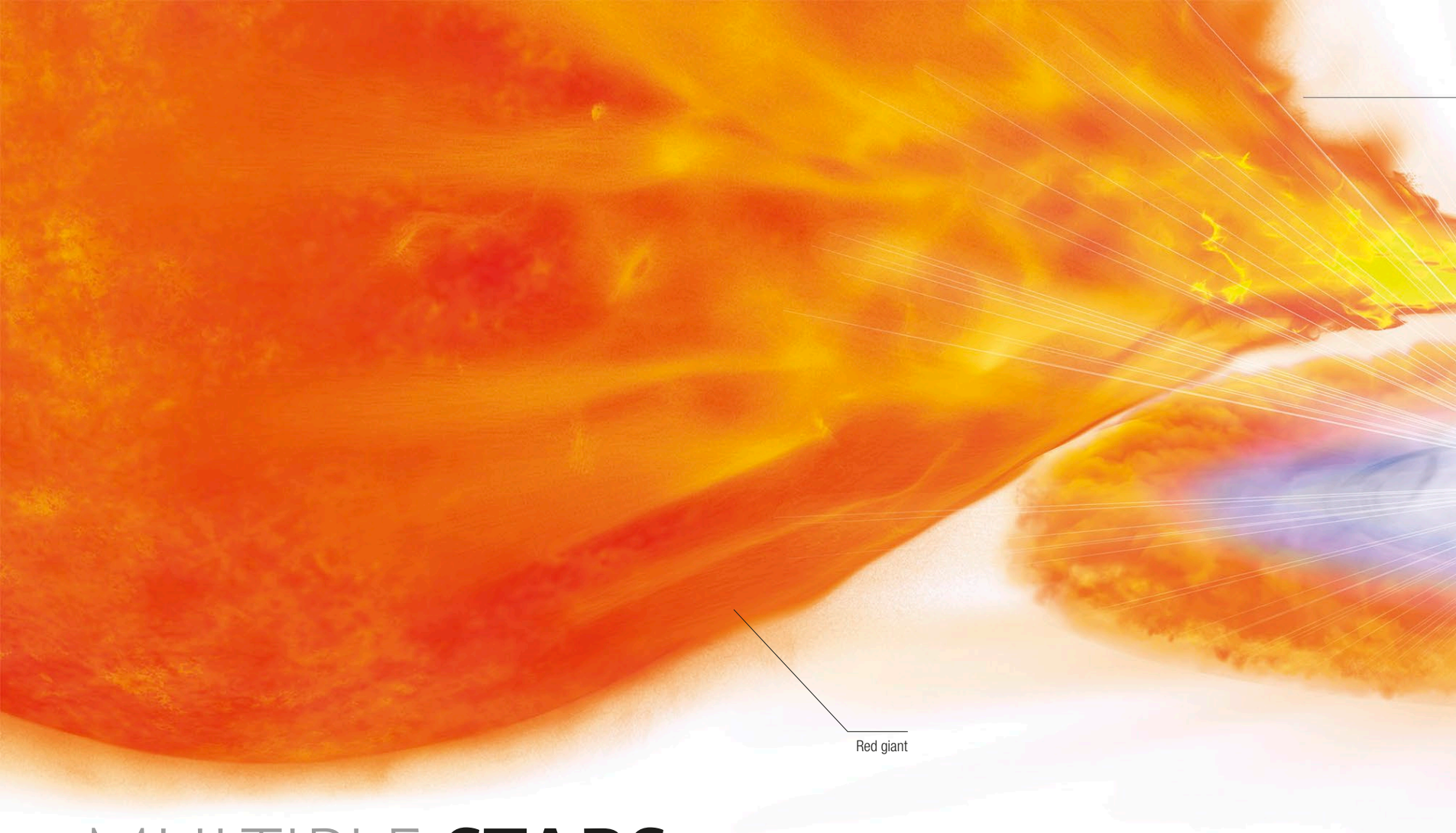




Red giant

MULTIPLE STARS

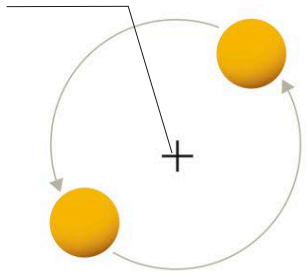
OUR SUN IS A LONE STAR WITH NO COMPANIONS. HOWEVER, MOST OF THE STARS WE CAN SEE IN THE SKY BELONG TO MULTIPLE-STAR SYSTEMS—THAT IS, TWO OR MORE STARS ORBITING EACH OTHER, BOUND BY GRAVITY.

The stars in a multiple-star system can orbit one another in various different ways. A pair of stars circling around a common center of gravity is called a binary system. If the two stars have the same mass, the center of gravity is halfway between them. More commonly, one star is heavier than the other, and the two stars have orbits of different sizes. In systems of three or more stars, various more complicated orbits are possible. For example, two stars may orbit each other closely, with the third circling the closely orbiting pair at a great distance. Overall, more than half the stars in the Milky Way Galaxy are part of multiple-star systems. These systems are different from star clusters (see pp.44–45), which are large collections of stars only loosely bound by gravity.

True and optical binaries

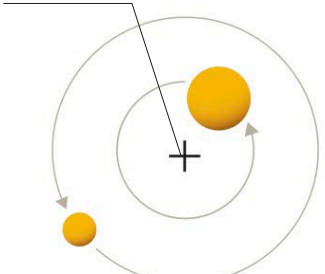
A star that looks like a single point of light may actually consist of two stars located very close together in the sky. Where these stars are also close together in space and gravitationally bound—they orbit each other—they are known as “true” binaries. An example is Albireo in the constellation Cygnus (see pp.124–25). In contrast, some star pairs that happen to be close in the sky are not close in space, and are not gravitationally bound—they just happen to be in the same direction as seen from Earth. Doubles of this sort are called optical doubles. An example is a star pair called Algedi, or Alpha Capricorni, in the constellation Capricornus (see pp.186–187). Its two components are more than 600 light-years apart.

Center of gravity
equal distance
from stars



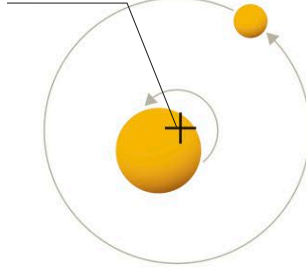
△ **Equal mass**
In binaries that consist of two stars of equal mass, the stars will orbit a common center of gravity, which lies midway between the two stars.

Center of gravity
closer to more
massive star



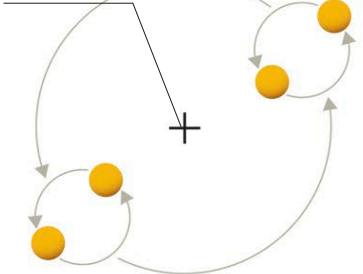
△ **Unequal mass**
If one of the stars in a binary system is more massive than the other, the system's center of gravity lies closer to the higher-mass star.

Center of gravity
lies inside
high-mass star



△ **Significant difference in mass**
Sometimes one star is much heavier than the other. In such cases, the center of gravity may lie at the surface of the more massive star, or even inside it.

Single center
of gravity



△ **Double binary**
In a double binary or quadruple system, each star typically orbits one companion, and the two pairs orbit a single center of gravity.